

Mebrahtu G. Tedla, Ph.D.

Research Scientist: Immunology & Translational Research
Indiana University School of Medicine | +1-573-902-8623 | mtedla@iu.edu



PROFESSIONAL SUMMARY

Translational immunologist with expertise in immune engineering, biologics delivery, cell-based assays, and preclinical disease models. Proven experience bridging mechanistic biology with therapeutic development in immunology, cell therapy, and inflammatory disease. Contributor to high-impact, cross-functional research programs including Global Burden of Disease studies published in *The Lancet* and *JAMA Network*.

CORE SCIENTIFIC SKILLS

- Immune Engineering & Tolerance Induction
- Biologics & Drug Delivery Platforms
- Cell & Gene Therapy Support (Islet Engineering)
- High-Dimensional Flow Cytometry & Immune Profiling
- Preclinical In Vivo Models (Mouse)
- Omics: Transcriptomics, Lipidomics, Metabolomics, Microbiome
- Data Analysis: R, Python, GraphPad, FlowJo

INDUSTRY-RELEVANT EXPERIENCE

Postdoctoral Research Fellow – Translational Immunology

Indiana University School of Medicine (2024–Present)

- Lead mechanistic studies identifying lipid-mediated immune pathways relevant to asthma and allergy therapeutics
- Apply flow cytometry, metabolomics, and in vivo models to generate actionable biomarker data
- Collaborate with clinicians and multidisciplinary teams to align discoveries with translational goals

Postdoctoral Research Fellow – Immunology

University of Missouri School of Medicine (2022–2024)

- Developed lipocoacervate-based sustained-release biologics for immune tolerance induction
- Supported cell therapy-relevant strategies in transplantation and autoimmune disease
- Executed immune functional assays and preclinical validation studies

SELECTED PUBLICATIONS (TRANSLATIONAL)

- Journal of Immunology (2024) – Islet surface engineering for immune protection
- STAR Protocols (2024) – Translational islet transplantation methodology
- International Journal of Molecular Sciences (2024) – Immune memory modulation
- The Lancet / Lancet Infectious Diseases (GBD Consortium, 2025)

EDUCATION

PhD, Immunology – University of Melbourne (2020)

MSc, Epidemiology – Institute of Tropical Medicine, Antwerp (2015)

MSc, Medical Biotechnology – University of Gondar (2013)

Doctor of Veterinary Medicine – Hawassa University (2009)

PROFESSIONAL AFFILIATIONS

American Association of Immunologists | American Society for Microbiology |
ASTMH

ADDITIONAL

Experienced in regulated research environments (IACUC, biosafety, HIPAA, GMP-adjacent workflows). Strong communicator with demonstrated success in cross-functional scientific teams.